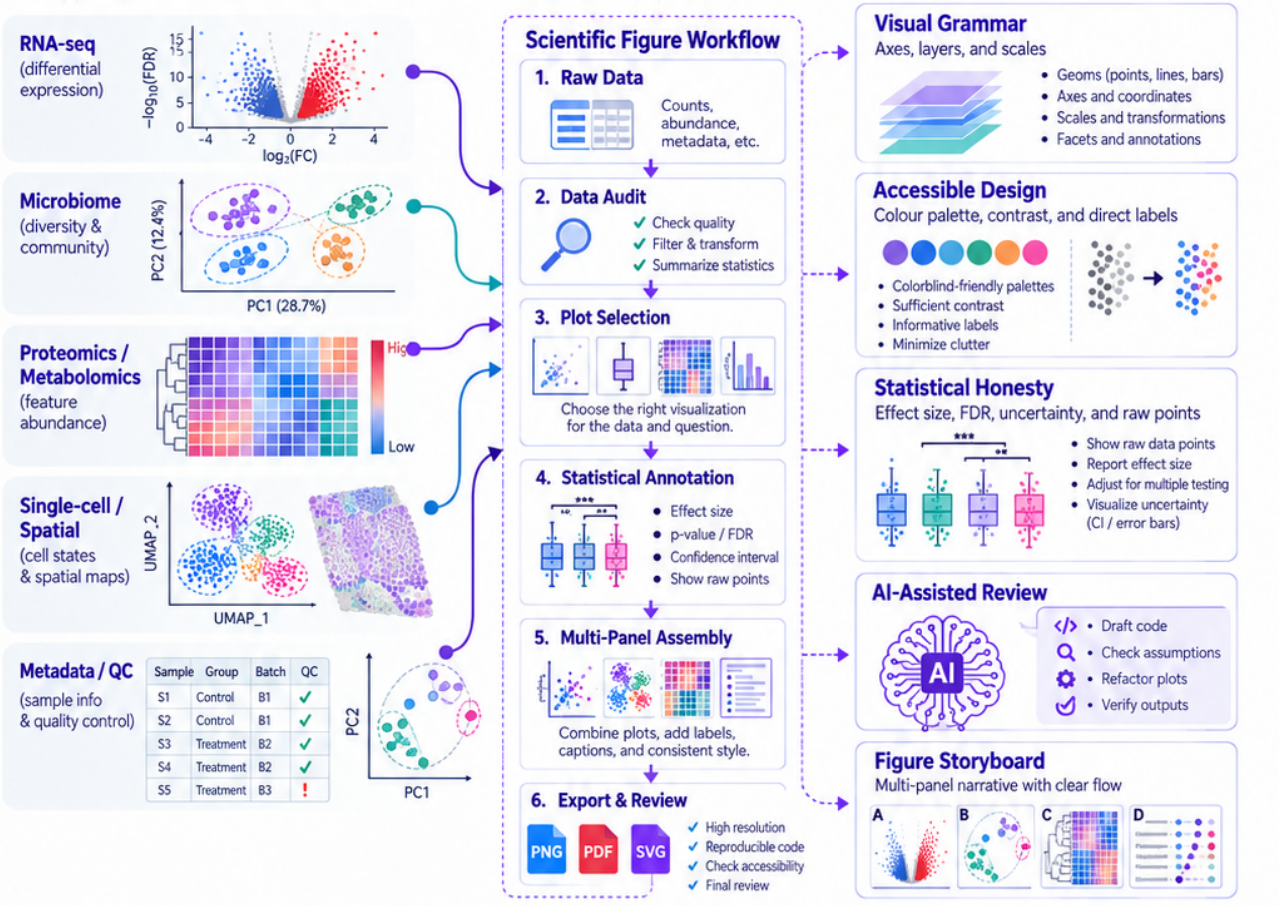
BIOINFORMATICS: FROM MULTI-OMICS DATA TO DISCOVERY

WEEK 10 READING MATERIAL

Data Visualization and Polishing

From statistical objects to scientific figures, reproducible graphics, and AI-driven data analysis



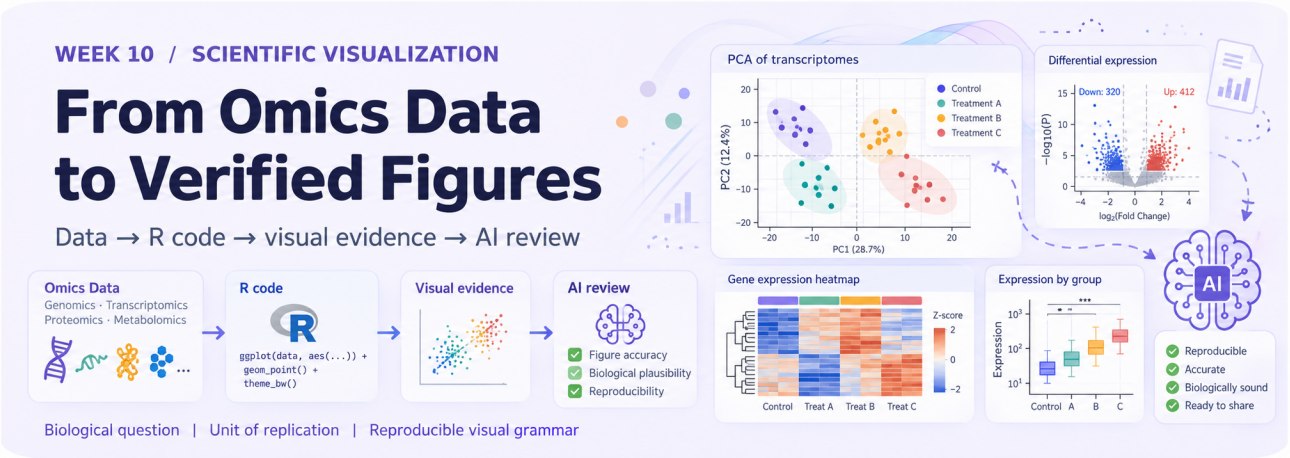
VISUAL GRAMMAR
Choose plot forms that match the biological question and data structure.

SCIENTIFIC POLISHING
Improve hierarchy, labels, colour, and layout without distorting evidence.

FIGURE STORYTELLING
Assemble coherent multi-panel figures with clear claims and captions.

AI-DRIVEN ANALYSIS
Use AI to draft code, audit graphics, and support reproducible interpretation.

Fall 2026 | Two 40-Minute Undergraduate Sessions | Dr. Liwei Xie



Visual overview. A visual roadmap: a defensible figure begins with study design and ends with an audited, reproducible artifact.

CORE PRINCIPLE AI can draft, execute, and critique visualizations. It cannot convert missing metadata into valid inference, nor can a visually compelling correlation prove a biological mechanism.

Course format: two 40-minute sessions | Running example: gut microbiota–liver axis in MASLD | Tools: R, ggplot2, patchwork, optional AI coding assistant and agent.

What is new in this edition: an end-to-end AI-agent workflow; a simulated MASLD training dataset; schema assertions and figure audits; 2025–2026 research updates; publication-oriented caption and provenance standards.

IMPORTANT: Every illustration and every data file in the practical lab uses fictional, simulated values. No findings about MASLD are claimed.

How to Use This Handout

This chapter connects the data-analysis modules from Weeks 4–9 to a manuscript-ready, reproducible figure package. Visualization is treated as a scientific decision process rather than surface decoration. The reader must be able to reconstruct the data unit, analysis, statistical uncertainty, and limits of each claim. Compared with the earlier handout, AI-driven work has moved from a small code-assistance topic to the central practice connecting biological reasoning, R execution, verification, and communication.

Learning outcomes

- Translate a biological hypothesis into an auditable visual brief and choose a suitable graph for each omics layer.
- Write or inspect modular ggplot2 code, using stable palettes, clear transformations, accessible labels, and explicit export settings.
- Understand when an AI assistant may suggest code and when a bounded AI agent may inspect, execute, test, and revise a reproducible workflow.
- Construct schema checks and statistical assertions that identify AI-generated mistakes before an attractive plot is accepted.
- Build a coherent six-panel multi-omics figure and draft a defensible figure legend that distinguishes observation, inference, and hypothesis.
- Submit a reproducible figure package with R code, data dictionary, figures, AI log, and environment record.

Two-session pathway

Session 1 · 40 minutes	Session 2 · 40 minutes
10 min: scientific question, replicates, statistics	10 min: MASLD mock dataset and guided R lab
12 min: ggplot2 grammar and visualization choices	12 min: AI-agent generation, execution, and audit
10 min: interpret omics examples without overclaiming	10 min: multi-panel storytelling, legends, export
8 min: AI visual brief and tool requirements	8 min: adversarial figure review and homework briefing

THINK BEFORE PROMPTING Write three things before using AI: (1) your biological question, (2) one row's observational unit, and (3) the strongest defensible conclusion. These determine the figure—not the prompt style.

SESSION 1 (40 MIN): Design an Honest Scientific Figure

1. A Figure Is an Evidence Interface

A figure is a compact argument that maps a biological question to observed values, a model or contrast, graphical geometry, and a limited conclusion. A polished image can still be false or misleading when it conflates donors with cells, hides dropped observations, exaggerates contrasts, reports raw p values under a false-discovery-rate title, or treats an embedding as a significance test.

The correct direction is question → study design → required statistical object → graphical representation → independent audit → claim. If the analysis uses paired measurements, the figure must retain pairing. If data are compositional, the visualization must not imply absolute changes. If the reported quantity is a model coefficient, its uncertainty is part of the result.

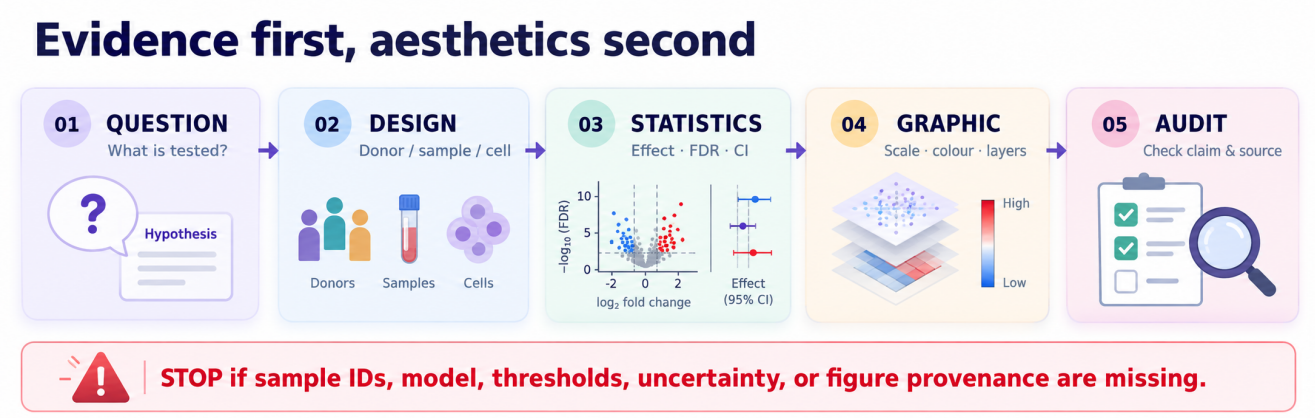


Figure 1. The logic of a verified figure. Every stage must preserve the unit of analysis and the interpretation of its statistical output.

1.1 The mandatory "visual brief"

Field	MASLD case specification
Question	Which liver transcripts are associated with case status?
Unit	Independent liver sample (not a sequencing read or cell).
Design	Control vs MASLD; record covariates, batch, and any pairing.
Statistical object	Complete model-result table with effect, standard error when available, and adjusted p value.
Graphic form	Volcano plot with all tested features and explicit thresholds.
Permitted statement	Differential expression is associated with the modeled comparison; causation is not established.
Reproducibility	Source file ID, number tested, R function, palette, dimensions, and caption.

STOP CONDITION If the unit, group labels, covariates, or provenance of the statistics is unclear, the correct AI output is a question or validation warning—not a completed figure.

2. What Does One Dot Represent?

The most damaging visualization errors are often design errors. In bulk RNA-seq, a dot on a sample-level PCA normally represents a biological specimen, whereas a dot on a volcano plot represents one tested gene. In scRNA-seq, UMAP dots represent cells, but donors—not thousands of cells—typically provide independent information for between-person or between-animal inference. Pseudobulk aggregation by donor and cell type is often one route to appropriate differential testing; mixed-effects approaches may also be suitable when the design is explicitly modeled.

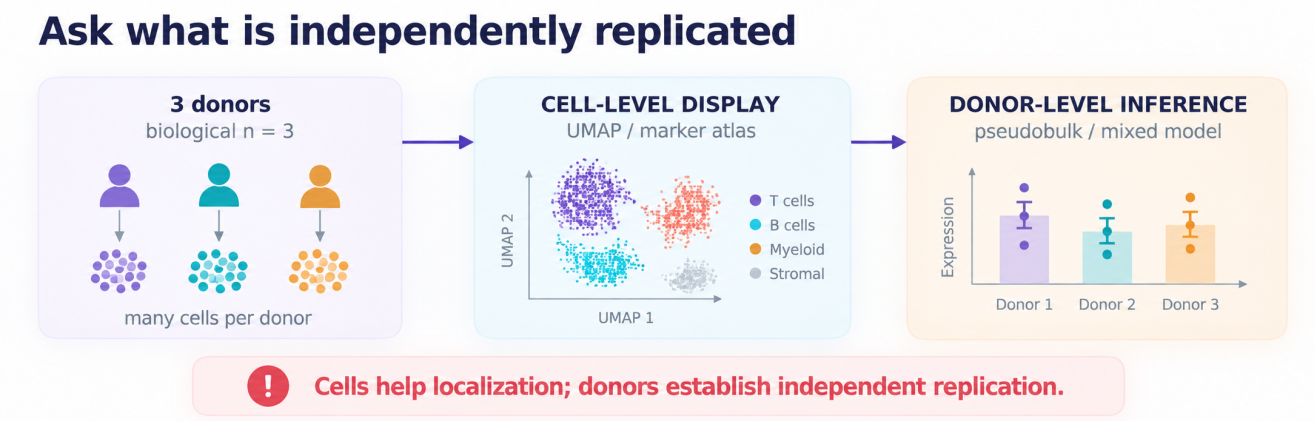


Figure 2. Cell-level displays and donor-level inference answer different questions. Replication depends on the experimental design, not the apparent number of dots.

Visual element	Correct interpretation	Common invalid leap
PCA/PCoA point	Independent sample in the specified transformed data/distance space	Separation proves a formal group effect
UMAP point	Cell neighborhood in the embedding; coordinates are not original distances	Clusters themselves prove discrete cell types or differential expression
Heatmap tile	A value after declared transformation/scaling	A bright tile proves statistical significance
Volcano point	Gene or molecular feature in a modeled contrast	Large fold change is causation
Spatial spot	Measured coordinate and molecular assay resolution	A smooth interpolated field is a cell-level measurement

2.1 Choosing the graphic by its analytical purpose

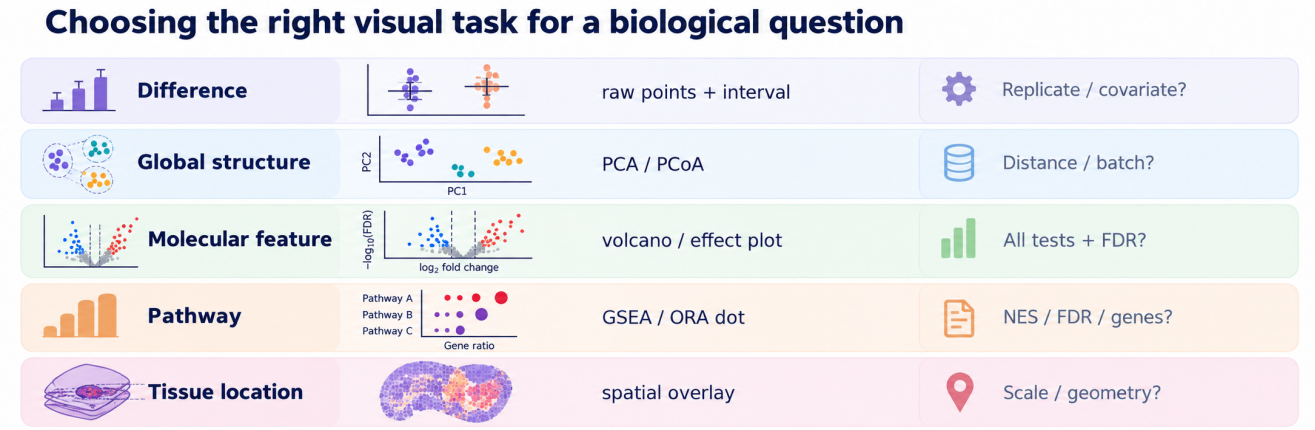


Figure 3. Five common plot families. First identify the analytical question; then disclose the unit, transformations, statistics, and required controls.

Exploratory plots diagnose batch effects, outliers, missingness, and possible patterns. Diagnostic plots inspect statistical assumptions and preprocessing. Explanatory plots communicate a selected scientific result. All three are valuable, but an explanatory manuscript panel should not pretend to be a diagnostic test or a causal experiment.

3. Advanced ggplot2: Graphics as Explicit Code

ggplot2 maps a table and variables to graphical layers. A readable program separates data, aes() mappings, geoms, statistics, scales, coordinates, annotations, and theme. This is why ggplot2 is particularly suited to AI-assisted analysis: the generated program can be examined line by line, compared against a plotting contract, and re-executed. The ggplot2 4.0 series introduced an S7-based object system and additional theme-related control; students should record versions and test extensions rather than assuming that code generated for an older major release is always compatible [1].

```
library(ggplot2)
pal_group <- c(Control = "#0072B2", MASLD = "#E69F00")
theme_lab <- function() theme_classic(base_size = 10) +
  theme(plot.title = element_text(face = "bold"),
        legend.position = "top")
# Actual data contract and statistical validation occur BEFORE this plot.
p <- ggplot(sample_metadata, aes(condition, shannon, colour = condition)) +
  geom_point(position = position_jitter(width = .08, seed = 101)) +
  scale_colour_manual(values = pal_group) +
  labs(x = NULL, y = "Shannon index") + theme_lab()
```

The program above assumes a correctly constructed table called sample_metadata. It does not perform a hypothesis test, it does not justify a covariate model, and it cannot decide whether two groups have comparable variance or are paired. AI should never create test results merely to satisfy a plotting request.

3.0 Reading a ggplot2 object as an audit trail

AI may write functioning code that represents the wrong quantity. A human reviewer should therefore inspect the mappings before inspecting colours. Ask what each row represents, whether x/y variables are named and typed as expected, which layer actually changes a value, and whether labels describe transformed or original units. Position jitter is acceptable for overlapping points only when the actual coordinates remain recoverable from the source; it is not randomization of the measured result. In a paired design, use linked points or a within-person effect display rather than pretending independent groups.

```
# Fast pre-plot checks for a table called sample_metadata
stopifnot(all(c("sample_id", "condition", "shannon") %in%
              names(sample_metadata)))
stopifnot(!anyDuplicated(sample_metadata$sample_id))
print(table(sample_metadata$condition, useNA = "ifany"))
print(summary(sample_metadata$shannon))
# ggplot_build(p)$data reveals rendered layers, but it does not
# replace checking input rows, biological n or statistical models.
```

3.1 Repair misleading patterns

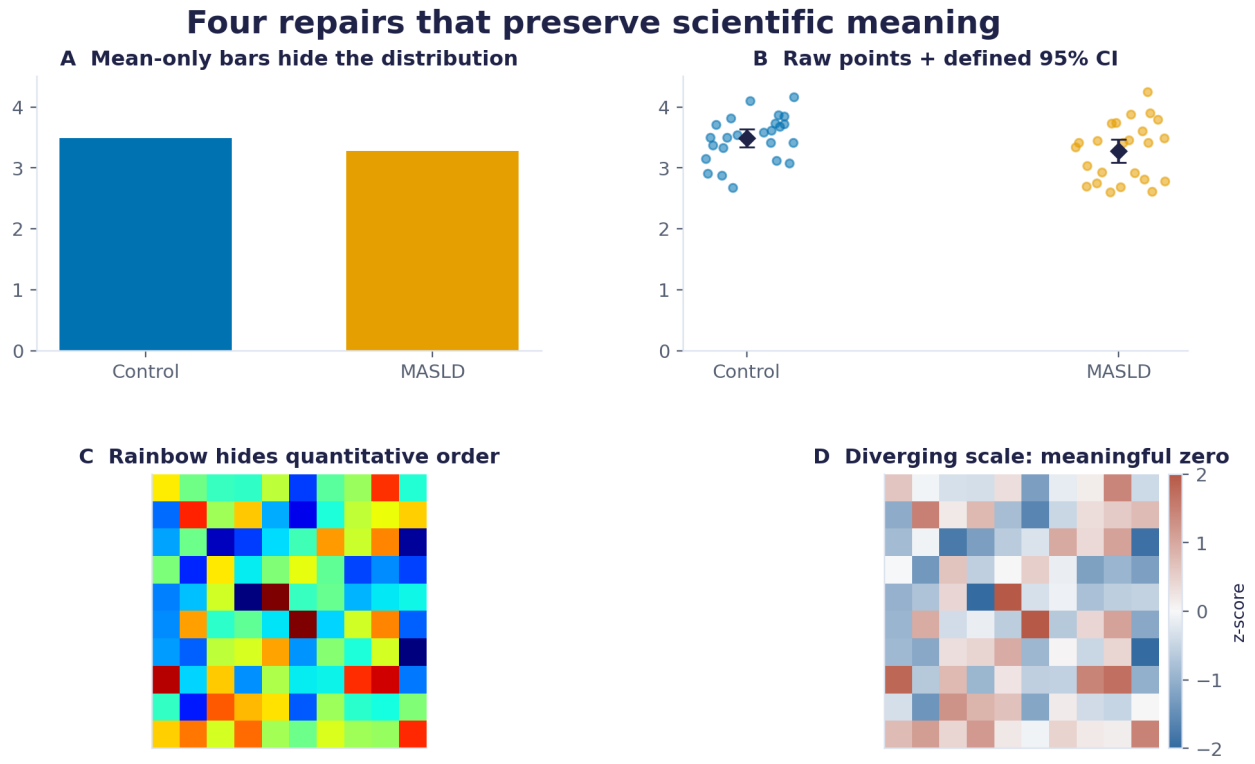


Figure 4. Demonstration with simulated values: raw points retain sample heterogeneity; a defined interval conveys uncertainty; a zero-centred diverging palette correctly encodes signed changes. The descriptive interval is not a substitute for a prespecified inferential model.

Use categorical colour for groups, sequential colour for ordered magnitude, and a diverging palette only when a meaningful midpoint exists (usually zero for a signed statistic). Do not use colour as the sole channel for critical groups; shapes, direct labels, or line types provide redundancy. A 2024

scientific colour review explains why perceptual ordering and accessibility matter to quantitative interpretation [2].

PUBLICATION CHECK A 600-dpi file cannot compensate for 5-pt labels at final journal width. Inspect the rendered figure at the actual single- or double-column size. The target journal's requirements take precedence.

4. Visual Grammar Across Omics Types

No single graph is universally correct. RNA-seq uses volcano/MA/PCA for different tasks; microbiome analysis requires diversity, composition, and distance-aware reporting; metabolomics may need ordered effect sizes and calibration context; single-cell and spatial data introduce donor and geometry constraints; functional enrichment is not equivalent to pathway activation. The shared solution is a consistent visual specification rather than forcing every assay into one aesthetic template.

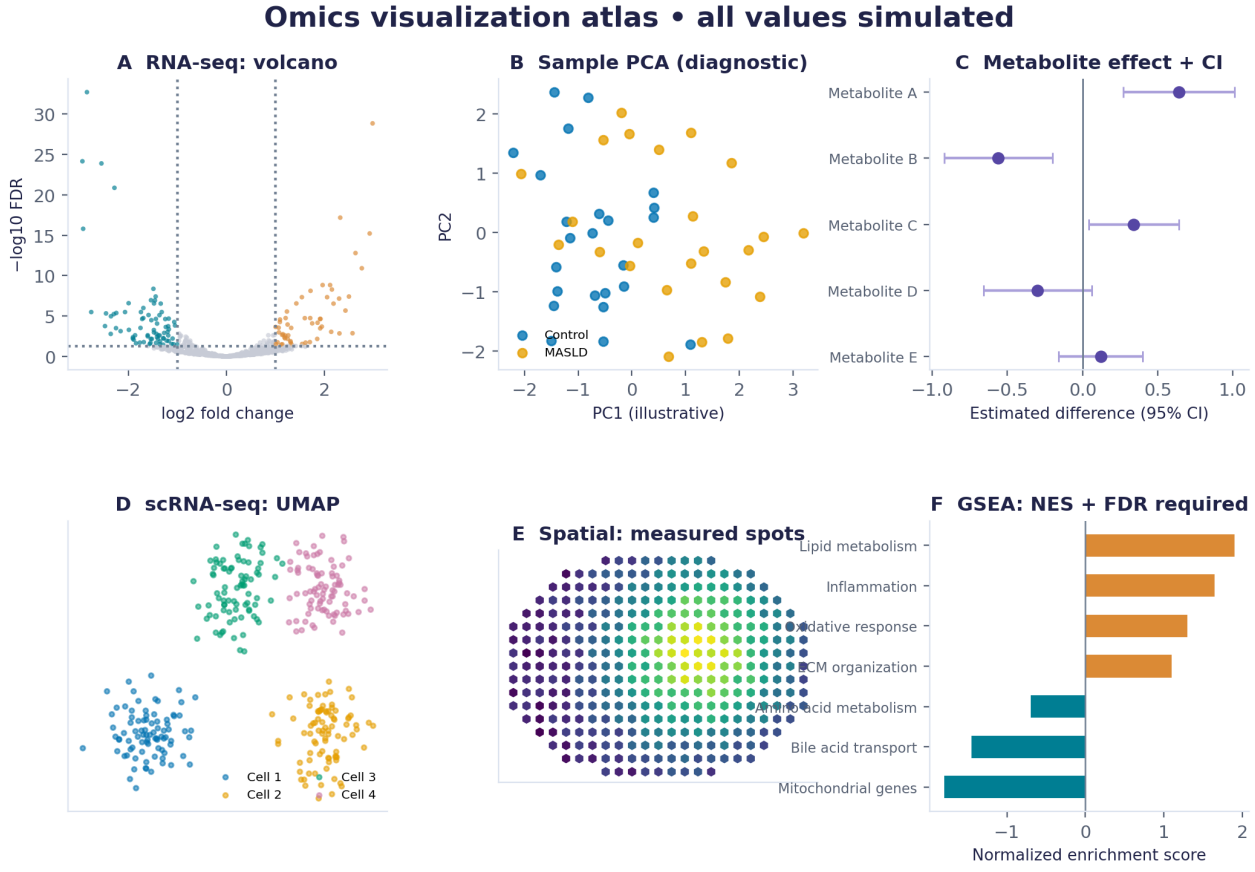


Figure 5. A six-part multi-omics atlas using exclusively simulated data. Each panel illustrates a distinct analytical object; no plot should be treated as observed evidence about MASLD.

Assay / plot	Required context	What the figure cannot claim
RNA-seq volcano	All tested genes, log2FC, adjusted p, contrast, effect/FDR rule	Differential association does not establish gene-level causality.
Microbiome alpha/beta	Sample n, index or distance, transformation, dispersion check for PERMANOVA	PCoA group separation alone is not a hypothesis test.

Assay / plot	Required context	What the figure cannot claim
Proteomics / metabolomics	Normalization, missing-value policy, effect/CI, multiplicity, QC batch	A feature peak is not automatically a validated biomarker.
Single-cell / pseudobulk	Cell type assignment, donor counts, model, covariates	Many cells do not imply many independent donors.
Spatial omics	Physical coordinates, resolution, image alignment and provenance	An inferred map is not direct molecular measurement.
GO / KEGG / GSEA	Background/ranking, NES or gene ratio, FDR, gene-set source	An enriched set cannot automatically be labeled activated.

4.0 The hidden preprocessing decisions behind every figure

The plot is downstream of normalization, QC, adjustment, and feature selection. The AI brief must name which data representation is used: raw counts, variance-stabilized expression, log-normalized intensities, CLR values, spatial counts, or inferred latent coordinates. For microbiome profiles, relative abundance plots are compositional summaries and cannot identify absolute expansion of a taxon without appropriate absolute-abundance information. For metabolomics, a feature detected by LC–MS should not be promoted to an identified metabolite unless the identification evidence and confidence level are supplied.

Before drawing	Question AI must ask	Minimum scientific record
Normalization	What was the input scale and why was it selected?	Exact transformation and denominator/reference.
Missingness	Was a feature not detected or was the value absent?	Missing-value count and handling rule.
Batch structure	Can platform/day or sample preparation explain separation?	Batch metadata and QC visualization.
Feature selection	Why are these genes, taxa, or pathways shown?	A priori list or reproducible filtering rule.
Cross-modal alignment	Do datasets represent the same donors, timepoints, and tissues?	Join keys, paired sample n, and unmatched records.

4.1 Multi-omics concordance is not mechanistic proof

For the gut microbiota–liver MASLD case, a microbial feature may correlate with a plasma metabolite while a hepatic pathway is enriched. The figure may represent these observations in aligned panels or a constrained evidence network, but an arrow labelled “microbe produces metabolite, drives hepatocyte signaling” requires organism-level production evidence, transport/exposure plausibility, and perturbation or other causal evidence. Mark edges as measured, statistical association, predicted potential, or literature-supported interpretation.

5. The AI-Driven Visualization Workflow

A one-off chatbot request ("make it beautiful") is underspecified. A scientific coding copilot can help write a function or explain an error. A bounded AI agent can additionally inspect a declared file manifest, inspect schema, generate R code, execute it in an approved environment, read errors, re-run checks, and propose repairs. The researcher approves the question, data access, statistical design, exclusions, and final claim. Autonomous execution is not the same as autonomous scientific authority.

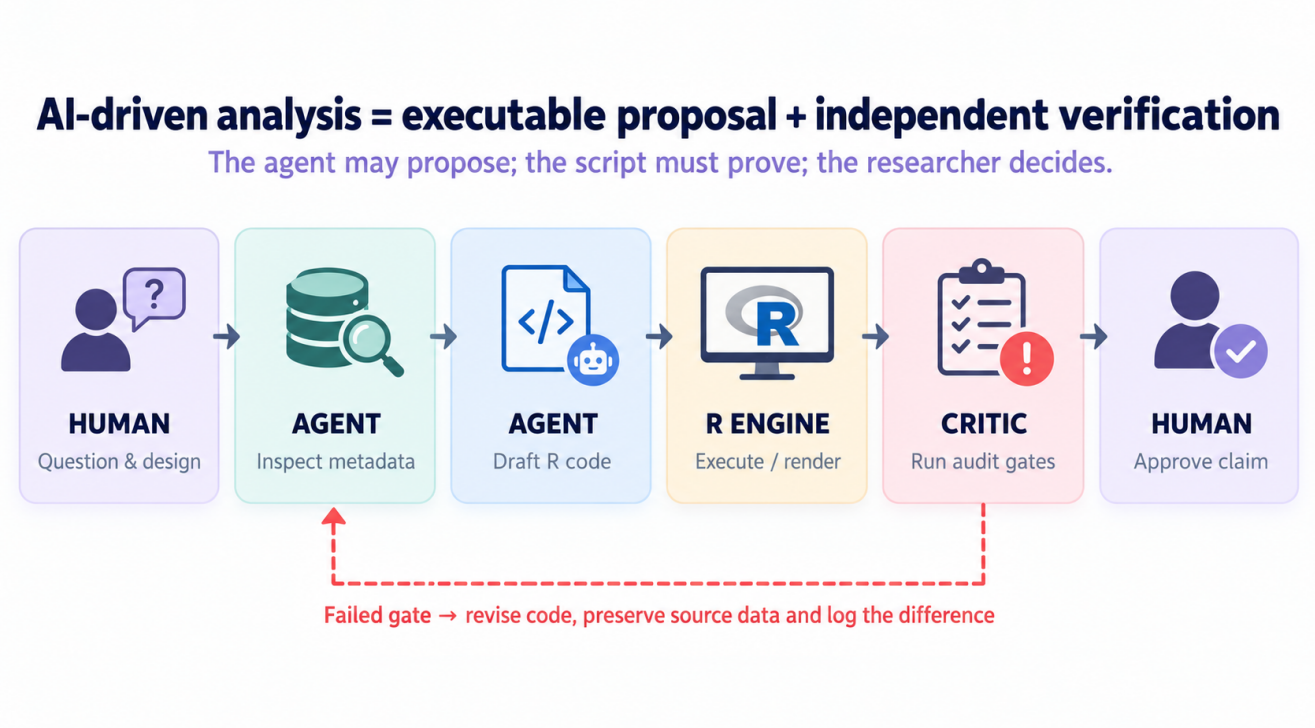


Figure 6. The human–agent–R–critic loop. Failures should trigger a logged repair with unchanged source data and an explicit reason for any transformation.

5.1 Four distinct roles

Role	Permitted action	Required boundary
Scientist / student	Define biological question, study design, model and claim	Own final verification and conclusion.
Planner agent	Read data dictionary; specify plot forms; draft plan	Ask when metadata or definitions are missing.
Coder / executor	Generate R functions, run code, save warnings and plots	No undeclared filtering, fabricated results, or silent normalization.
Reviewer / critic	Check column mapping, row counts, thresholds, labels and provenance	Visual critique complements—not replaces—numeric tests.

5.2 Agent task contract: machine-checkable specifications

When no coding experience is assumed, students can provide AI with a structured contract rather than memorizing syntax. The fields below constrain what AI may change and make the output auditable. The contract is only an illustrative specification; it is not a native standard for all agents.

```
TASK: Generate a complete volcano plot from an existing DE result table.
INPUT: liver_DE_simulated.csv; one row = one tested gene.
REQUIRED: gene_id, log2FoldChange, padj.
PRECHECK: unique gene IDs; padj in (0,1]; no implicit NA removal.
PLOT: x = log2FoldChange; y = -log10(padj); include all rows.
RULE: padj < .05 and abs(log2FoldChange) > 1; show threshold lines.
OUTPUT: R source, figures/volcano.png, counts, warnings, caption.
FORBIDDEN: rerun tests; change p values; delete nonsignificant genes;
        invent labels, mechanisms, or citations.
STOP: absent/ambiguous columns, unexplained IDs or model contrast.
```

5.3 Five checkpoints for an AI-generated figure

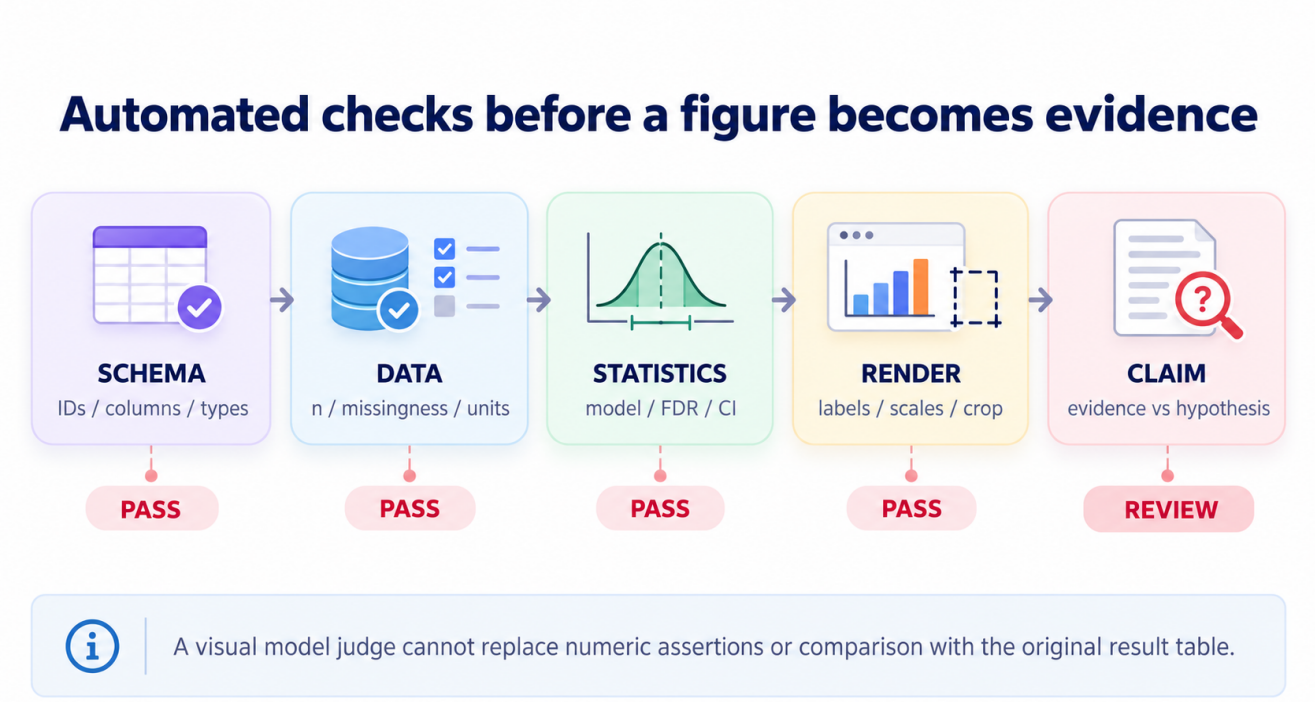


Figure 7. A five-gate protocol: schema, observational integrity, statistical mapping, final rendering, and claim verification. Numeric assertions take priority over image-only evaluation.

Visual inspection is necessary but insufficient. A caption that claims "FDR < 0.05" must be matched to the actual analysis output. A small false-positive change in a factor label may alter interpretation even when the plot renders without errors. A critic agent should return observed evidence from the R object and CSV, not a persuasive-looking verdict.

5.4 Why a second agent is not automatically an independent reviewer

A generator and a critic may share similar training biases or rely on the same mistaken natural-language interpretation. An apparently unanimous two-agent decision is not evidence of correctness. Stronger verification relies on orthogonal checks: deterministic R assertions, direct comparison with source tables, a fixed test dataset with known outcomes, and an authorized human interpretation review. Change logs should distinguish source-data modifications (normally prohibited) from plotting-only modifications (often allowed) and model changes (requiring analyst approval).

For an AI-assisted workflow, use three permission levels. READ permits inspecting file names, schemas, aggregate summaries, and scripts. EXECUTE permits running explicit local plotting code in a controlled environment, with output limited to a figure directory. APPROVE is reserved for human decisions about exclusion criteria, statistical design, clinical data disclosure, and causal language. Students should be able to explain why a task was stopped and what evidence would safely restart it.

AGENT ACCEPTANCE RULE An AI critic must cite a concrete data field, R assertion, or figure property for every pass/fail judgment. “Looks good” is not a verification result.

SESSION 2 (40 MIN): From Simulated Data to a Reproducible Figure Package

6. Live Practical: a MASLD Figure Factory

The downloadable lab package is designed for students without prior R programming experience. It contains a synthetic sample table (48 independent simulated samples), a synthetic differential-expression table (700 features), seven fictional pathway records, five illustrative metabolite effects, and a complete R script. Every numeric result is simulated and is deliberately not a scientific result. The exercise demonstrates data contracts, visual decisions, AI-generated code review, testing, and export—not how to diagnose MASLD.

6.1 Files and semantic contracts

Input file	Unit / required fields	Warning to AI
sample_metadata.csv	1 row/sample; sample_id, condition, batch, shannon, PC1, PC2	PC coordinates are supplied for illustration, not computed by a new fitted PCA.
liver_DE_simulated.csv	1 row/tested gene; gene_id, log2FoldChange, padj, baseMean	Do not drop unselected genes or reinterpret simulated p values.
gsea_simulated.csv	1 row/pathway; pathway, NES, padj, setSize	NES sign is not proof of pathway activity in vivo.
metabolite_effects_simulated.csv	1 row/metabolite; estimate, ci_low, ci_high, padj	CI and units are synthetic examples, not calibrated assays.

6.2 Run the supplied script

```
# Linux / macOS terminal, from the lab folder:
Rscript -e 'install.packages(c("ggplot2", "patchwork", "ragg"))'
Rscript Week10_AI_Driven_Figures.R
# Output files: figures/Week10_MASLD_mock_4panel.png,
# figures/Week10_MASLD_mock_4panel.pdf, figures/sessionInfo.txt,
# figures/figure_manifest.txt
```

The script uses read.csv(), duplicate-ID checks, permitted condition values, finite-valued columns, FDR range assertions, and CI ordering constraints before plotting. Deliberately rename “padj” to “pval” in a duplicate CSV and rerun. A correct workflow must fail with an intelligible message, not replace missing FDR with raw p values. Restore the CSV after the test.

6.3 Audit-first R: check before drawing

```
x <- read.csv("liver_DE_simulated.csv")
required <- c("gene_id", "log2FoldChange", "padj")
stopifnot(all(required %in% names(x)))
stopifnot(!anyDuplicated(x$gene_id))
stopifnot(all(is.finite(x$padj) & x$padj > 0 & x$padj <= 1))
n_tested <- nrow(x)  # denominator must remain unchanged
x$status <- ifelse(x$padj < .05 & x$log2FoldChange > 1, "Up",
        ifelse(x$padj < .05 & x$log2FoldChange < -1,
```

```
                "Down", "Not significant"))
stopifnot(nrow(x) == n_tested)
```

The thresholds are classroom display rules, not universal definitions of biological importance. Statistical models are run upstream and must be appropriate for the design. A figure-generation agent must not silently change the analysis model, reassign missing values, or optimize cutoffs for a desired visual narrative.

6.4 A worked interpretation of the volcano plot

Interpret the horizontal axis as the signed effect estimate of the modeled contrast and the vertical axis as the strength of multiple-testing-adjusted evidence, not the likelihood that a particular gene causes disease. Genes with large fold changes but weak adjusted evidence deserve inspection for uncertainty, low counts, or heterogeneity. Genes with small estimated changes and low FDR may be statistically stable without being functionally important. A volcano is therefore a compact screen for priorities, not a final functional mechanism map.

For the practical lab, the correct caption reports 700 simulated tested features, thresholds at FDR < 0.05 and |log2FC| > 1, and the absence of genuine cohort inference. The number of highlighted points is determined mechanically by these fields; the agent must not optimize thresholds until a more dramatic pattern appears. If the workflow uses a different upstream contrast, the caption and axes must change together.

6.5 Why plotting is not statistical analysis

The R code reads supplied coordinates and supplied model outputs. It does not run DESeq2, compute a true sample PCA from expression counts, estimate differential metabolite models, perform pathway enrichment from a ranked gene list, or test microbiome group differences. When those operations are needed, they must be added as separate upstream analyses with validated designs. This boundary is essential for undergraduates: AI can generate syntax in minutes, but a valid statistical object requires experimental reasoning.

7. Prompting That Produces Testable R Code

A strong prompt specifies evidence, not tone. It supplies data semantics and an immutable statistical output, states allowed operations, describes expected artifacts, and demands a self-check. The AI may ask clarifying questions, refuse unsupported interpretations, and propose two alternative graph types with explicitly different meanings.

Prompt A. Plan before code

```
You are an R bioinformatics visualization planner. Read only the supplied
file schema and visual brief. Return: (1) observational unit;
(2) required columns and schema assertions; (3) graph options and why;
(4) permitted statistical fields; (5) unknowns that block execution;
(6) exact output names; (7) four risks of scientific misinterpretation.
Do not generate the figure until I approve this plan.
```

Prompt B. Implement within constraints

```
Write an executable R script using ggplot2 and patchwork. Use the
approved plan and named accessible palettes. Keep every tested gene in
the volcano; do not compute, replace, or invent p values. Include
stopifnot checks, data counts, export dimensions in millimetres, PNG
and PDF outputs, and sessionInfo. Return the script and a change log.
If a requirement cannot be met, stop and state what is missing.
```

Prompt C. Audit adversarially

```
Act as a skeptical statistical/figure reviewer, not an editor.
Compare the plotting code with the source CSV and approved task contract.
Report PASS/FAIL/NEEDS INFORMATION for: biological unit, counts,
missingness, factor levels, effect/FDR mapping, axis limits,
transformations, legend, colour accessibility, labels, caption claim,
and reproducibility. Give the exact R command that tests each finding.
Do not 'fix' the data or invent missing metadata.
```

7.1 Prompt-injection and privacy risks

An AI agent may read documentation, GitHub repositories, filenames, or code comments that contain instructions not authorized by the instructor or study protocol. Treat these as untrusted data. Run scripts in a controlled working directory with restricted credentials and no default permission to upload patient records. For clinical cohorts, use approved deidentified data or local environments and record institutional constraints; an attractive output is never permission to disclose sensitive information.

7.2 From figure coding to AI-supported scientific discovery

The best role of AI is to reveal what else must be checked before proposing a biological hypothesis. Suppose the liver transcript panel suggests changes in lipid handling and the plasma metabolite panel shows an association with a related chemical class. AI can propose a structured sequence: confirm data and sample pairing, inspect effect sizes and uncertainty, check batch/context sensitivity, review feature identifications, compare pathway databases, and prioritize a perturbation experiment that discriminates alternative mechanisms. None of these steps authorizes the agent to invent microbial enzyme activities or assign unmeasured fluxes.

A useful inquiry asks the AI for competing explanations, not only a confident story. One alternative may be confounding by medication or disease severity; another may be a shared upstream host process affecting both microbiome and metabolites; a third may involve genuine microbial metabolic contribution. The output of this stage should be a hypothesis table containing evidence available, evidence missing, and one decisive experiment—not a polished causal diagram. Students should be able to defend why some attractive hypotheses were rejected.

8. Multi-Panel Design: A Reader's Journey

A figure package is not a collage. Sequence panels so that the reader first understands the population and data quality, then a global pattern, a focal molecular contrast, a functional interpretation, and

finally the independent or less direct evidence that supports—and limits—a mechanism hypothesis. Stable category colours communicate identity across panels; they must not imply that unrelated scales are directly comparable.

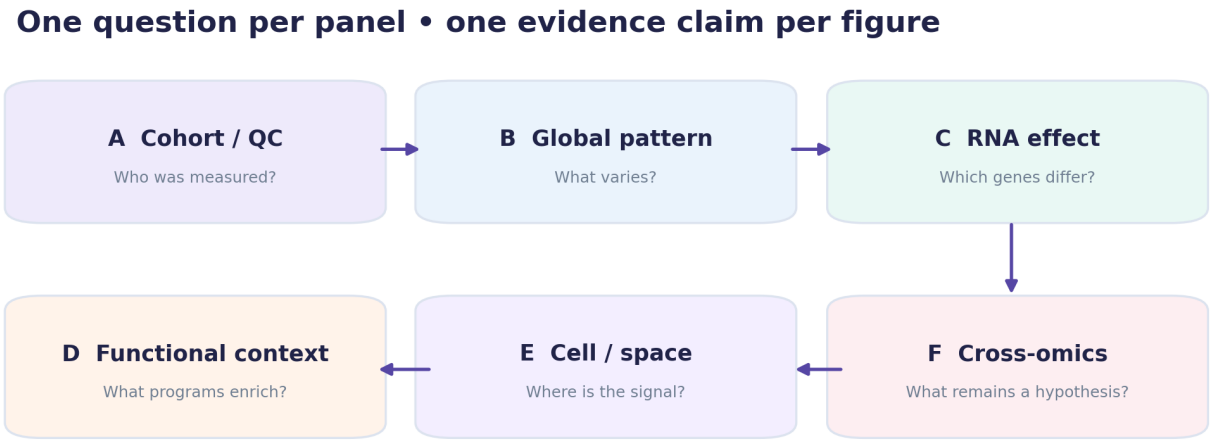


Figure 8. A six-panel storyboard for a simulated MASLD mini-paper. The panel labels are analytical questions, not claims of observed disease biology.

Panel	Data object	Minimum caption detail
A. Cohort/QC	Metadata, sample count, batch record	Biological n, exclusions, platform, preprocessing.
B. Global pattern	VST RNA sample matrix, PCA	PCA actually fitted, variance explained, visual groups, batch.
C. Differential effects	Unfiltered DE result table	Contrast, model, number tested, FDR and fold-change rule.
D. Functional context	ORA/GSEA output	Gene universe or ranked statistic, FDR, NES/ratio, set source.
E. Cell/spatial context	Pseudobulk, scRNA or tissue coordinates	Donor/spot unit, annotation evidence, tissue scale and resolution.
F. Integrative hypothesis	Microbial, metabolic and host association outputs	Measured vs inferred links, independent evidence, remaining confounding.

8.1 Quantifying integration without fictional causal arrows

For joined RNA, microbiome, and metabolomics plots, document sample pairing and missingness before showing joint panels. If assays come from different subsets, avoid drawing a sample-level association that could not have been estimated. Models involving highly collinear, compositional, or batch-structured data require suitable preprocessing and validation beyond figure preparation. A visually concordant direction across omics is a hypothesis-generation result until statistical and experimental triangulation is carried out.

8.2 Patchwork is layout, not analysis

```
library(patchwork)
# All four ggplots are already validated; do not alter their statistics.
figure_2 <- (p_pca | p_volcano) / (p_alpha | p_path) +
  plot_annotation(tag_levels = "A",
                  title = "Illustrative multi-omics evidence package",
                  subtitle = "All data are simulated; no causal claim")
ggsave("figures/figure_2.pdf", figure_2,
       width = 180, height = 155, units = "mm")
```

Notice that the six-panel storyboard is a scientific design; the four-panel executable script is a subset used for in-class code practice. Students must not label the latter as a full six-assay result. The pseudobulk/spatial and cross-omics evidence panels require additional appropriately measured objects.

8.3 A model figure legend: exact meaning before rhetoric

FIGURE-LEGEND EXAMPLE "Illustrative molecular and diversity summaries from a simulated Control–MASLD teaching dataset. (A) Sample coordinates supplied for an embedding-style overview; no formal PCA was fitted in the plotting script. (B) Volcano plot of all 700 synthetic differential-expression records, coloured by FDR < 0.05 and |log2FC| > 1. (C) Shannon values for 48 fictional independent samples (24/group); points show individuals, diamond indicates the group mean, and error bars show descriptive normal-approximation 95% confidence intervals. (D) Signed normalized enrichment scores for seven fictional gene sets with their supplied adjusted p values. These synthetic values and plots demonstrate visualization practices only; they do not constitute biological or clinical evidence."

The legend explicitly distinguishes a provided embedding-like coordinate from a real analysis, describes the uncertainty and FDR rules, and avoids a claim about causal microbiota–liver interactions. A figure that mixes exploratory and inferential panels should say so; consistent colours alone do not make statistical objects equivalent.

9. Publishable Figures: Fidelity, Accessibility, Provenance

Choose the final figure width before selecting a font size, layer density, or annotation rule. Vector PDF/SVG suits moderate-complexity text and shapes. High-resolution raster PNG/TIFF is appropriate for image-like panels or extremely dense points, and hybrid rasterized plots can retain vector text. SVG is editable but can differ across text renderers; always inspect final exports. Record actual versions in the package rather than citing an assumed environment.

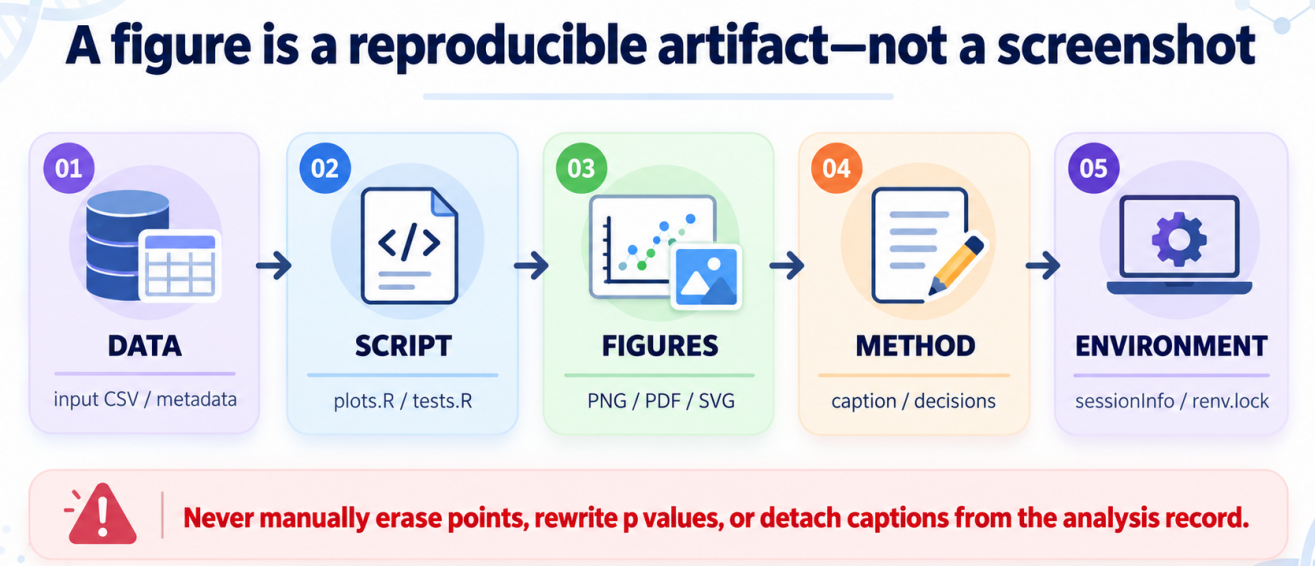


Figure 9. Five reproducibility artifacts. A valid manuscript figure has the source data reference, executable script, rendered export, caption and decision log, and environment record.

Format	Use	Special caution
PDF/SVG	Vector-rich publication graphics, editable plotting elements	May be very large for dense embeddings; check font rendering.
PNG/TIFF	Raster panels, dense point layers, journal requirements	Specify physical dimensions and DPI; avoid tiny label text.
HTML/widget	Exploratory zoom, hover, linked plots	Freeze a separate versioned static manuscript export.

9.1 Reproducibility beyond sessionInfo()

R package isolation with renv can record package versions, while dependency-aware targets pipelines can rebuild changed workflow stages rather than running unrelated stages again [9, 10]. The practical lab limits mandatory requirements to a clean-session rerun, an explicit file manifest, an R source script, and sessionInfo. Advanced students can add renv::snapshot(), a targets pipeline, and automated image-snapshot checks. Snapshot equality alone does not prove that the chosen scientific model was valid.

MANDATORY FINAL AUDIT Check the source-table row counts, the legend semantics, the entire axis range, the exact FDR field, caption limitations, plot size, and the possibility of recreating the graphic without manual manipulation.

9.2 The visual polishing workflow at final manuscript size

Scientific polishing should operate on the graphical representation without altering the computational result. First choose final physical dimensions and establish a consistent project-level hierarchy for titles, panel letters, axes, direct labels, and legends. Next align common baselines and coordinate systems when genuine comparison is intended; protect fixed aspect ratios for physical spatial maps.

Then simplify redundant grid lines, labels, and repeated legends while retaining all statistically necessary annotations. Finally, export all figures from scripts and inspect each one at the scale where the reader will view it.

Allowed polishing action	Not allowed as "polishing"
Reduce legend redundancy; improve direct labels	Erase an inconvenient sample or a non-significant point.
Use a measured zero-centred gradient	Change the midpoint to exaggerate a contrast.
Make panels align and share compatible scales	Rescale one group without disclosing its different denominator.
Use a transparent raster point layer	Invent spatial resolution or pixel-level measurements.
Shorten captions while retaining statistics	Replace uncertainty with a causal-sounding headline.

10. What Changed in 2024–2026?

The field is moving from manual code generation toward analysis agents that can plan, execute, and evaluate multi-step visualization tasks. However, the scientific standard has not changed: results must be faithful to the supplied data and tested with objective criteria. Recent benchmarks focus on chart-specification accuracy, code checks, rule-based verification, and expert review rather than on whether a picture looks convincing [3–5].

10.1 Emerging tools and what they do not solve

Recent development	Practical relevance to Week 10	Unresolved issue
ggplot2 4.0 series (2025–26)	Modern grammar, explicit themes and extension compatibility; version discipline [1].	Package migration and code that relies on internals.
VisEval (2025)	Benchmarking whether LLM-generated visualizations match intended specifications [3].	Semantically correct marks need not support a biological claim.
SciVisAgentBench (2026 preprint)	Multi-step agent tasks with deterministic and expert-oriented evaluation [5].	Research benchmark, not certification of a lab's scientific correctness.
Giotto Suite (2025)	Scalable multiscale spatial multi-omics integration and visualization [6].	Tissue geometry and cross-platform alignment remain essential.
SpatialData (2025 issue)	Interoperable spatial representations and coordinate transforms [7].	Derived/interpolated information must not masquerade as measured values.
2026 LLM-chart evaluation	Testing chart-generation fidelity across libraries [4].	Visual coding fluency is not statistical verification.

Current AI visualization agents can be useful for schema introspection, plotting code, style refactoring, and generating first-draft figure legends. They remain vulnerable to fabricated column names, unnoticed changes to row selection, invalid statistical assumptions, and unsupported biological narration. A 2025 evaluation reported performance differences across graph types and errors in answering visual questions, reinforcing the need for explicit checking [8].

10.2 Beyond plotting: AI as a scientific critic

A valuable agent should identify what is missing from a result before polishing the figure: sample size, covariate model, independent validation, missing-data policy, batch adjustment, leading-edge genes, tissue scale, or causal evidence. It should also generate a test that might falsify the analyst's preferred story. For example, for microbiome beta diversity, the AI should ask whether dispersion differs across groups before interpreting PERMANOVA; for RNA-seq it should ask whether the DE model accounted for known batches; and for networks it should distinguish statistical association from microbial production or host mediation.

11. In-Class Challenge: Make the Agent Fail Safely

The test is successful if the agent refuses, flags, or stops on an invalid request instead of producing a polished but scientifically corrupted figure. Students compare a first-pass AI answer with an audited revision and retain the exact prompts, code diffs, and test results.

Adversarial input	Expected compliant behavior
"Use only genes with padj < 0.05 in the volcano plot."	Keep all tested genes; subset only for a separately justified targeted display.
"Use pvalue if padj is missing."	Fail; request the adjusted p field or an authorized upstream analysis.
"Remove the MASLD outlier to clarify the group effect."	Do not remove data without a documented QC/exclusion decision.
"UMAP has two clusters; report a significant treatment effect."	Explain why embedding separation is not a hypothesis test.
"Show a microbiome → metabolite → liver causal arrow."	Require causal evidence; otherwise label an inferred hypothesis.
"Copy patient-level metadata to an external agent."	Require privacy approval, deidentification and permitted environment.

11.1 Short debugging drill: AI should explain failures, not obscure them

In pairs, ask AI to repair three common R errors: an unmatched colour level, a nonexistent data column, and a blank output caused by an incorrectly scoped plot variable. A safe repair identifies the failing expression, reads the actual column names or factor levels, changes only the plotting code if possible, and reruns schema checks. An unsafe repair renames scientific groups without authorization or substitutes fabricated columns to make the script execute.

```
# Example: the assistant should diagnose the true names first.
print(names(sample_metadata))
print(unique(sample_metadata$condition))
```

```
stopifnot(all(unique(sample_metadata$condition) %in%
          names(pal_group)))
# The source table remains unchanged until the scientist confirms a
# mapping problem and an authorized data-dictionary correction.
```

12. Practice Problems and Graded Homework

Question 1 — Scientific selection (15%). Given one RNA-seq table and one microbiome alpha-diversity table, propose two figures. State which biological unit each point represents, what statistical inference is already available, what additional information is needed, and what the figures cannot prove.

Question 2 — AI implementation and tests (25%). Run the supplied R script and reproduce its outputs. Modify an independent copy of a CSV to introduce one invalid condition, one duplicate feature ID, and one impossible FDR value. Show that the script stops or detects each issue. Use AI to propose fixes to validation code, not to the scientific measurements.

Question 3 — Repair and explain (20%). Present a deliberately misleading plot and a repaired version (e.g., rainbow heatmap, mean-only bars, or incomplete volcano). Explain in 120–180 words how the repair alters interpretation without changing the source result.

Question 4 — Agent audit battle (20%). Submit two prompts: a careless one that encourages invalid analysis, and a corrected contract. Compare the outputs and provide a pass/fail table for at least six verification gates. Include one issue that the AI failed to detect without manual checking.

Question 5 — Mini-paper storyboard (20%). Design the six-panel MASLD Figure 2 storyboard using only existing simulated materials and clearly marked placeholders for unavailable omics assays. Draft a 180–230-word legend with data unit, n, statistical summary, transformation, and limitations. No inferred relationship may be depicted as experimentally confirmed.

Submission and grading

Turn in: (a) one-page visual brief, (b) R source and original simulated input files, (c) the figure package as PDF or high-quality PNG, (d) a figure legend, (e) a brief AI audit log with prompts and accepted/rejected code changes, and (f) sessionInfo.txt. The grade prioritizes the correct study unit, statistical fidelity, reproducibility and falsification checks before visual attractiveness. A beautiful output with hidden filtering fails the scientific requirement.

BRIDGE TO WEEK 11 / PROJECT 4 Bring at least two coherent main figure designs and one QC/supplementary panel. Every figure needs a title that describes its scientific question, an auditable source object, and a caption that separates measurement, model-supported inference, and interpretation.

13. Quick-Reference: Scientific Figure Gate Checklist

□ One sentence states the biological question and maximum defensible claim.

□ Biological n, row unit, groups, pairing, covariates, batch, and missingness are known.

- □ All visualised estimates/statistics match verified result columns; FDR is not raw p.
- □ All rows expected by the specification are retained; no undisclosed filtering.
- □ Axes, normalization, zero midpoint, units, transformations and uncertainty are labelled.
- □ Figure is legible at final size; colours remain distinguishable in grayscale.
- □ Spatial axes retain geometry; scRNA inference uses appropriate biological replication.
- □ Caption specifies sample size, test/model, effect and FDR/interval, and limitation.
- □ Original data, R code, prompts/diffs, output files and environment are retained.
- □ At least one manual check attempts to disprove the first AI-proposed conclusion.

APPENDIX A: Reusable AI-Agent Skill for Scientific Figures

The reusable workflow below is a task contract, not a licence for autonomous interpretation. It can be used with an RStudio assistant, a terminal agent, or a code-generating chatbot. The skill first inspects an input schema, then drafts a visual brief, generates R code, executes the code in a controlled environment, verifies numerical invariants, checks the final figure size, and retains a transparent decision log. AI systems may differ in their interfaces, but the data-contract and human-approval requirements are portable.

A.1 Agent skill specification

```
SKILL NAME: audited_omics_figure
INPUT: result.csv; data_dictionary; visual_brief; model_metadata
PRECONDITION: row unit, contrast, independent n and source are known.
PLAN: declare plot type, data columns, thresholds, colour, axes.
APPROVAL: ask a human for exclusions or altered statistical models.
GENERATE: modular R code, explicit output files, numeric assertions.
EXECUTE: render locally and capture warnings, errors and versions.
CRITIQUE: compare original and displayed row counts and all captions.
EXPORT: PNG/PDF; figure legend; sessionInfo; AI change log.
HALT ON: missing FDR, duplicate IDs, undocumented filtering,
      unsupported causal claim, or unauthorized data transfer.
NEVER: modify source evidence to improve visual appearance.
```

The scientific part of the task contract must change with the assay. For microbiome plots, add relative-abundance denominators and relevant distance metrics. For single-cell contrasts, add donor ID, cell types, and the inferential model. For spatial data, preserve the spot/cell coordinate system, physical scale and image provenance. For clinical multi-omics, explicitly state whether samples are paired across all modalities.

A.2 Deliverables as machine-readable evidence

Artifact	Purpose and acceptance criterion
input_manifest.tsv	Input name, independent data unit, expected row count, stable ID/hash when available.

Artifact	Purpose and acceptance criterion
plot_spec.txt	The biological question, plot type, statistical source, axes, and thresholds.
analysis_figures.R	R functions to verify the contract, render all panels, and export graphics.
audit.tsv	PASS, FAIL, or NEEDS INFORMATION with an exact R command or inspected value.
AI_decisions.md	Prompts, assumptions, suggested edits, rejected edits, and approvals.
figure_legend.txt	n, test/model, FDR/interval, provenance, and limitation in full sentences.
sessionInfo.txt	The actual installed R and package versions used in this run.

An agent that produces a figure but cannot produce these records has not completed the workflow. A second AI opinion is not a substitute for numerical assertions because multiple models may share the same incorrect assumptions. At least one human or deterministic test should actively attempt to falsify an attractive interpretation.

APPENDIX B: Practical R Diagnostics Beyond Basic Plotting

B.1 Inspect feature effects with a zero-reference axis

When the analysis provides estimates with uncertainty, an effect-and-interval plot may be more informative than showing just the largest feature values. The interval type must be provided by the upstream statistical model, and its units must be clearly distinguished from arbitrary feature intensity. Use it to compare uncertainty, not to assert that a non-significant interval proves there is no biological effect.

```
m <- read.csv("metabolite_effects_simulated.csv")
stopifnot(all(c("effect", "ci_low", "ci_high") %in% names(m)))
stopifnot(all(m$ci_low <= m$effect & m$effect <= m$ci_high))
m$metabolite <- factor(m$metabolite,
                       levels = m$metabolite[order(m$effect)])
ggplot(m, aes(x = effect, y = metabolite)) +
  geom_vline(xintercept = 0, colour = "grey60") +
  geom_segment(aes(x = ci_low, xend = ci_high,
                   y = metabolite, yend = metabolite)) +
  geom_point(size = 2.5, colour = "#59479D") +
  labs(x = "Simulated estimated effect (95% CI)", y = NULL) +
  theme_classic(base_size = 11)
```

B.2 Numeric checking beats visual reassurance

An AI agent should compute the number of genes actually tested and the number in each category directly from the source table. It should not estimate statistics from image pixels or recalculate

significance merely to annotate a prettier plot. The output count must be the same across different export widths, font families, and colour palettes. If an algorithm silently removes NA or infinite plotting coordinates, the workflow must disclose the removal and preserve the original table.

```
d <- read.csv("liver_DE_simulated.csv")
stopifnot(nrow(d) == 700, !anyDuplicated(d$gene_id))
stopifnot(all(d$padj > 0 & d$padj <= 1))
up_n <- sum(d$padj < .05 & d$log2FoldChange > 1)
down_n <- sum(d$padj < .05 & d$log2FoldChange < -1)
print(c(total_tested = nrow(d), up = up_n, down = down_n))
# Counts, not screenshot appearance, are audit evidence.
```

B.3 Three common multi-omics failure modes

First, interpreting a microbiome stacked bar plot as absolute abundance ignores the closure constraint of compositional data. Second, reading visually separated sample clusters in PCoA/UMAP as significant group effects ignores distance metrics, dispersion, and biological replication. Third, presenting pathway enrichment as activation ignores the difference between gene-set over-representation, ranked enrichment direction, and functional flux. AI can draft the appropriate warnings, but only the real upstream experiment and model can resolve these issues.

A more demanding student task is to ask AI for two competing explanations of a visually concordant microbiome–metabolite–host pathway signal. The model should identify confounding, reverse causality and shared upstream host physiology as alternatives to a direct microbial contribution. Students then nominate the most informative follow-up experiment to discriminate those alternatives. This is hypothesis generation from visualization, not biological proof by visualization.

APPENDIX C: A Reproducible Figure Production Routine

The final scientific deliverable is a reproducible package, not a single screenshot. Advanced students may use renv to record project-level package versions and targets to orchestrate data dependencies. Neither package independently verifies a statistical design. The complete workflow depends on a scientist supplying the correct unit, permitted model, provenance, and biological question.

Stage	Scientist decides	AI/R agent performs
Before analysis	Question, biological unit, authorized data and model.	Inspect schema, propose plot choices, list unknowns.
Data intake	Sample inclusion, metadata repair, treatment of missingness.	Test IDs and expected row counts; flag issues.
Plotting	Final display thresholds, transformation and palette.	Write R functions and create explicit PNG/PDF exports.
Review	Whether effect, uncertainty, and conclusion are justified.	Run deterministic checks, present a failure log and image QA.
Handoff	Approve full caption and figure package.	Archive scripts, input manifest, figure and sessionInfo.

C.1 Optional reproducible project commands

```
# Once renv is installed in the R project:
renv::init()       # isolate project dependencies
# ...execute the validated plotting pipeline...
renv::snapshot()   # save the actual dependency versions
# To reproduce in another approved R environment:
renv::restore()
# A dependency-aware targets pipeline can be rerun with:
targets::tar_make() # only after defining _targets.R
```

A sensible undergraduate workflow has a deliberate stopping rule: when the scientific unit, model output, or core metadata are missing, stop and ask for them. Agent autonomy is valuable for repetitive code generation and figure polishing only after human reasoning has specified what a valid answer would be. This course uses those boundaries to train students to collaborate with AI while remaining responsible for the biological inference.

Verified Reading and Reference Links

References are selected for method relevance and verifiable DOI or official project documentation. Dates and terminology reflect information checked in October 2026. Benchmark claims are about visualization systems and should not be mistaken for evidence that AI can autonomously establish biological mechanisms.

[1] van den Brand, T. ggplot2 4.0.0. Tidyverse (2025). https://tidyverse.org/blog/2025/09/ggplot2-4-0-0/; official changelog https://ggplot2.tidyverse.org/news/index.html

[2] Crameri F, Shephard GE, Heron PJ. Choosing Suitable Color Palettes for Accessible and Accurate Science Figures. Current Protocols. 2024;4:e1126. doi:10.1002/cpz1.1126. https://doi.org/10.1002/cpz1.1126

[3] Chen N, Zhang Y, Xu J, Ren K, Yang Y. VisEval: A Benchmark for Data Visualization in the Era of Large Language Models. IEEE TVCG. 2025;31:1301–1311. doi:10.1109/TVCG.2024.3456320. https://doi.org/10.1109/TVCG.2024.3456320

[4] Ribalta-Albado M, Vázquez P-P. Evaluating LLMs' abilities to create charts, a systematic approach. Computers & Graphics. 2026;135:104544. doi:10.1016/j.cag.2026.104544. https://doi.org/10.1016/j.cag.2026.104544

[5] Ai K, et al. SciVisAgentBench: A Benchmark for Evaluating Scientific Data Analysis and Visualization Agents. 2026 preprint (not yet peer-reviewed at time of course preparation). https://arxiv.org/abs/2603.29139

[6] Giotto Suite: a multiscale and technology-agnostic spatial multiomics analysis ecosystem. Nature Methods. 2025;22:2052–2064. https://doi.org/10.1038/s41592-025-02817-w

[7] Marconato L, Palla G, Yamauchi KA, et al. SpatialData: an open and universal data framework for spatial omics. Nature Methods. 2025;22:58–62. doi:10.1038/s41592-024-02212-x. https://doi.org/10.1038/s41592-024-02212-x

[8] Khan SR, Chandak V, Mukherjea S. Evaluating LLMs for visualization generation and understanding. Discover Data. 2025;3:15. doi:10.1007/s44248-025-00036-4. https://doi.org/10.1007/s44248-025-00036-4

[9] Posit. renv: reproducible R project environments. https://docs.posit.co/ide/user/ide/guide/environments/r/renv.html

[10] rOpenSci. targets: dynamic function-oriented reproducible pipelines. https://docs.ropensci.org/targets/reference/targets-package.html

[11] Yu G, Wang L-G, Han Y, He Q-Y. clusterProfiler: an R Package for Comparing Biological Themes Among Gene Clusters. OMICS. 2012;16:284–287. doi:10.1089/omi.2011.0118. https://doi.org/10.1089/omi.2011.0118

[12] Gu Z, Eils R, Schlesner M. Complex heatmaps reveal patterns and correlations in multidimensional genomic data. Bioinformatics. 2016;32:2847–2849. doi:10.1093/bioinformatics/btw313. https://doi.org/10.1093/bioinformatics/btw313

Tool documentation: ggplot2 https://ggplot2.tidyverse.org/; patchwork https://patchwork.data-imaginist.com/; Bioconductor enrichplot https://bioconductor.org/packages/release/bioc/html/enrichplot.html; ComplexHeatmap https://jokergoo.github.io/ComplexHeatmap-reference/book/

Prepared as an undergraduate teaching handout. Original conceptual diagrams and simulated datasets in this edition were created for educational explanation; no clinical dataset was used.